

CS 6603: AI, Ethics, and Society
Homework Project #3: AI/ML Part 1

Readings:

- Dixon, Lucas & Li, John & Sorensen, Jeffrey & Thain, Nithum & Vasserman, Lucy. “Measuring and Mitigating Unintended Bias in Text Classification,” AAAI/ACM Conference on AI, Ethics, and Society, pp. 67-73, 2018. https://www.aies-conference.com/2018/contents/papers/main/AIES_2018_paper_9.pdf
- Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai, “Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings,” NIPS 2016 -<https://papers.nips.cc/paper/6228-man-is-to-computer-programmer-as-woman-is-to-homemaker-debiasing-word-embeddings.pdf>

In this assignment, you’ll continue the process of exploring relationships in data. You’ll accomplish this task by computing some basic inferential statistical measures on a natural language-based dataset.

Natural language processing is concerned with the ability to process and analyze large amounts of natural language data, whether for automated sentence completion in emails, conversational agents and chatbots, or AI tools to help journalists. In this assignment, we will work with a sample of the data from a classifier built to identify toxicity in comments from Wikipedia Talk Pages. The model is built from a dataset of 127,820 Talk Page comments, each labeled by human raters as toxic or non-toxic. A toxic comment is defined as a “rude, disrespectful, or unreasonable comment that is likely to make you leave a discussion.”

Step 0: Toxicity data

Download the modified dataset available on CANVAS, called “*toxicity_per_attribute.csv*”:

- <Wiki_ID> is a unique identifier associated with Wikipedia comment
- <TOXICITY> is a toxicity value from 1 (if the comment was considered toxic) and value 0 (if the comment was considered neutral or healthy)
- <subgroup> columns: 1 column per human attribute; True if the comment mentioned this identity.

As you complete the steps below, please make sure that the *Steps* correspond to the sections in your Assignment. (ex. Step 1 should match to section 1 in your assignment).

Step 1: Explore the data

1. Explore and review the dataset. Determine whether any data cleanup is needed. If yes, outline the cleanup procedures you applied and report any resulting modifications to the data (e.g., changes in the number of records/observations or variables). If no, provide your reasoning for not performing cleanup.
2. Identify any protected classes and its subgroups.
Note: How the subgroups are classified is up to you. For example, subgroups, Christian and Muslim, could belong to the protected class Religion. Similarly, if a subgroup falls under two protected classes, it is up to you to decide how you want to classify it (i.e. keep it under both protected classes or keep it under one protected class). For example, Chinese could fall under Race or National Origin or both.

In your report, provide a table that lists the protected class categories and their corresponding subgroup members.

Note: These will be the protected classes and subgroups you will be using for the remaining steps.

Step 2: Calculating Toxicity Correlations

1. Create a *reduced data set* by deleting any rows that have all FALSE values for every column in that row (i.e., comments that reference no identity group).

Note: This is the *reduced data set* that will be used in all subsequent steps.

2. Using the reduced data set, identify an ordering scheme for each protected class by defining values for each of its subgroups. To do this, convert FALSE values to 0 and TRUE values to a unique numerical value for each subgroup member.

How the numerical values are assigned is up to you to design. For example, you could assign increasing integer values to subgroups within a protected class based on your belief of toxicity: Example gender: FALSE = 0; male = 1; female = 2; binary = 3.

You may also combine group members and assign numerical values based on your belief about similarities among the group members. This could be a possible mapping for gender: FALSE = 0; all others = 1; female = 2.

Another way is to calculate the mean toxicity of all the rows that are TRUE for a specific subgroup (ex. “male”). Then, determine the objective ordering based on the mean toxicity for each subgroup. For example, for sex, if the “male” average is 0.25, the “female” average is 0.26, then male is denoted by “1”, and the female is denoted by “2”.

In your report, provide the mapping of your assigned numerical values to its protected classes’ subgroups. Please provide a short explanation of your encoding rationale and its limitations.

3. Using your assigned numerical values, create a compacted data set by combining the subgroup into one column representing the protected class category (e.g. combine all columns related to Religion into one Religion column).
4. Calculate the correlation between the protected class columns and TOXICITY. In a table, provide the correlation coefficients for each protected class and the strength of the correlation.

Note: As guidance, you can use (Evans, J. D. (1996). *Straightforward statistics for the behavioral sciences*. Brooks/Cole Publishing) which suggests the following related to the absolute value of the correlation coefficient:

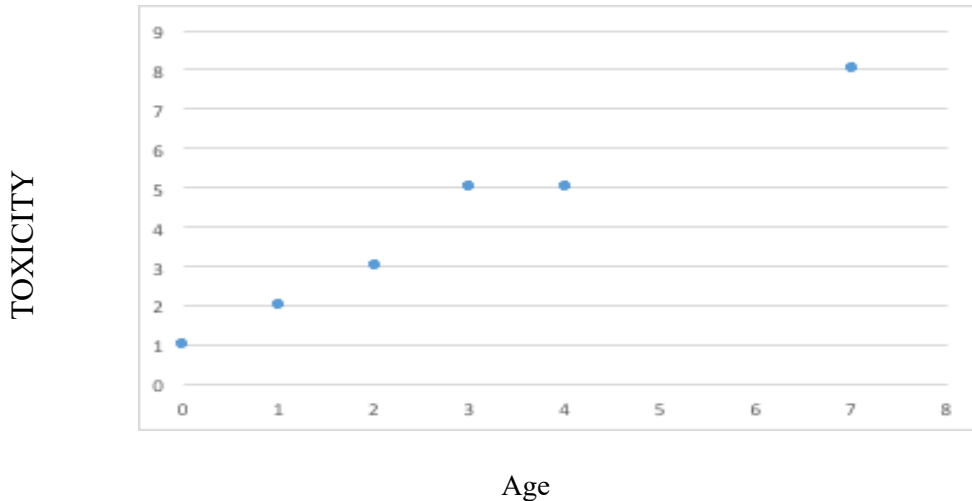
- .00-.19 “very weak” correlation
 - .20-.39 “weak” correlation
 - .40-.59 “moderate” correlation
 - .60-.79 “strong” correlation
 - .80-1.0 “very strong” correlation
5. Using the mappings from Step 2.2, plot the numerical Subgroup values vs Toxicity for the three highest correlation coefficients. Graphs should be a scatter plot similar to the illustrative graph provided in the example below.
 6. In your report, answer the following: Do the observed correlations meaningfully describe the relationship between comments that mention a subgroup and their toxicity? What are the limitations of using correlation in this context?

Step 2 Example Output:

Note: This example is for illustrative purposes only, your results might not match.

Correlation Coefficients:

	Religion (Protected Class Variable)	Age (Protected Class Variable)
TOXICITY	0.03	0.7
CORRELATION STRENGTH	Very weak correlation	Strong correlation



Step 3: Analyzing Toxicity on the Reduced Dataset

1. Using your reduced dataset (from Step 2.1), calculate and report the population mean and population standard deviation of TOXICITY. List the range of values around the mean that includes 95% TOXICITY.
Note: For this and the following questions, TOXICITY may/may not follow a normal distribution. Please use the methods discussed in the lecture (i.e. it is not expected for you to explore new methods).
2. Run the random sampling method using 10% of the data. Report the mean, standard deviation, and margin of error (assume 95% confidence).
3. Run the random sampling method using 60% of the data. Report the mean, standard deviation, and margin of error (assume 95% confidence).

IMPORTANT: For the next steps below, choose a protected class category for analysis.

Step 4: Analyzing Toxicity for a Chosen Protected Class

1. Using the reduced dataset (from Step 2.1), calculate and report the mean and standard deviation of TOXICITY associated with the protected class.
Hint: For this step, TOXICITY values should only be included in the calculation when the associated protected class value is not FALSE.

2. Using the 10% randomly sampled dataset (from Step 3.2), calculate and report for the protected class:
 - Mean
 - Standard deviation
 - Margin of error
 - Indicate (Yes/No) if protected class sample mean lies within the margin of error (computed using the protected class population statistics from Step 4.1)
 - Explanation of why it does or does not lie within the margin of error
3. Repeat the previous step using the 60% randomly sampled dataset (from Step 3.3).

Step 5: Analyzing Toxicity for the Subgroups of the Chosen Protected Class

1. Using the reduced data set (Step 2.1), calculate and report the mean and standard deviation of TOXICITY associated with each subgroup that is a member of the protected class selected in Step 4.
Hint: For this step, TOXICITY values should only be included in the calculation when the associated protected class value is not FALSE.
2. Using the 10% randomly sampled dataset (from Step 3.2), calculate and report for **each** subgroup of the protected class:
 - Mean
 - Standard deviation
 - Margin of error
 - Indicate (Yes/No) if subgroup's sample mean lies within the margin of error (computed using that subgroup's population statistics from Step 5.1)
 - Explanation of why it does or does not lie within the margin of error
4. Repeat the previous step using the 60% randomly sampled dataset (from Step 3.3).

Step 6: Plots and Toxicity Analysis

Plot on one graph the following:

1. the computed population mean and standard deviation (Step 3)
2. the computed mean and standard deviation for the protected class category (Step 4)
3. the computed mean and standard deviation for each subgroup of the protected class category (Step 5).

Answer the following questions. Justify your reasoning.

1. Which subgroup has the highest TOXICITY value? Why might this be the case?
2. Which subgroup has the lowest TOXICITY value? Why might this be the case?
3. Which subgroup has the largest difference in TOXICITY value when compared to the population mean? Explain why you think this group has the largest difference.
4. What type of human or systemic bias may be present in this dataset?
 - How can this bias be quantified?
 - How might it be reduced in future datasets or models?

Step 7: Submission

Submit a PDF report documenting your outputs for each question.

- **The report must follow JDF formatting.**
 - You can find a link to the JDF template here: [JDF Templates](#). The Microsoft Word template is recommended to ensure proper formatting and styling.
 - Reports that do not follow JDF formatting, or that are not neat and well organized, may receive up to a 10% deduction, regardless of content quality.
- **Name your file exactly as follows: GTuserName_Assignment_3.pdf**
 - *Example: gBurdell3_Assignment_3.pdf*
 - Deductions may be applied if the file name is incorrect or does not follow this convention.
- **All charts, graphs, and tables must be generated using Python, Excel, or another suitable non-AI software application.**
 - Points may be deducted for charts or tables generated using AI tools.
- If you used a Jupyter Notebook (.ipynb) for your work, you may submit it as a supplementary file. However, note that **PDF report is required** and will be used for grading.